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MIL-P-17574D(YD)
21 June 1991
SUPERSEDING
MIL-P-17574C(YD)
10 November 1983

MILITARY SPECIFICATION

PUMP-ENGINE SYSTEM, CENTRIFUGAL, WATER-GASOLINE-FUEL OIL, SKID-BASE-MOUNTED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single-stage centrifugal pump-engine systems. Systems have the pump and the engine q ounted on a common steel skid base, and are compact, self-contained, and readily movable.

1.2 Classification. The pump-engine system is of the following type:

TYPE

Type II - Diesel-engine-driven.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the specification (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4320

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SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage (Vehicular, Ignition, Lighting, and Starting).

MILITARY

MIL-P-10603 - Pumps and Pumping Units, Centrifugal, Power-Driven, for Water; Packaging of.

(Unless otherwise indicated, copies of specifications are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government Duplications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN PETROLEUM INSTITUTE (API)

API STD 610 - Centrifugal Pumps for General Refinery Service.

(Application for copies should be addressed to the American Petroleum Institute, 1220 L Street, N.W., Washington, DC 20005.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B 16.1 - Cast Iron Flanges and Flanged Fittings.
ASME B 16.24 - Bronze Pipe Flanges and Flanged Fittings, Class 150 and 300.
ASME B 40.1 - Gauges - Pressure Indicating Dial Type - Elastic Element.
ASME B 73.1M - Horizontal End Suction Centrifugal Pumps for Chemical Process.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

HYDRAULIC INSTITUTE (HI)

Standards of Hydraulic Institute.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, 14600 Detroit Ave., Cleveland, OH 44107.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J537 - Storage Batteries.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Unless otherwise specified (see 6.2), the pump-engine system shall consist essentially of the following: a single-stage centrifugal pump, with the pump casing split horizontally (axially) or vertically at the option of the manufacturer unless either type of casing is definitely specified; diesel engine for driving the pump; engine controls; and a steel skid base.

3.2 First article. When specified (see 6.2), the contractor shall furnish a system for first article inspection and approval (see 4.2.1 and 6.6)

3.3 Standard commercial product. The system shall, as a minimum, be in accordance with the requirements of this specification and" may be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, may be included in the system being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements of manufacturer's catalogs, or brochures, and represents the latest production model.

3.3.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied,

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance of maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice and shall be subject to all provisions of this specification. In accordance with

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ASME B 73.1M or API STD 610, as applicable, material and components, parts, and accessories shall be compatible with liquid hydrocarbon products including aviation gasoline, jet aircraft engine fuels, fuel oils, kerosene, and diesel oils. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new.

3.5 Interchangeability. All systems of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Pump design and construction. The pump-engine system shall be designed to withstand the strains, jars, vibrations, and other conditions incidental to shipping, storage, installation, and operation. Unless otherwise specified (see 6.2), the pump shall have flanged connections in accordance with ASME B 16.1 for class 125/250 and ASME B 16.24 for class 150/300, with a 6-inch intake and a 4-inch discharge connection. As required, flanged type eccentric reducers and concentric increasers shall be provided. Discharge and intake sides of the pump shall be conspicuously identified with arrow markings indicating flow, and at least one of the sides, at the manufacturer’s option, shall be conspicuously identified by the word "DISCHARGE" or “INTAKE”, as applicable. Symbols shall be readily visible and cast integrally with the casing unless traced by welding bead, cut into the casing or inscribed on a firmly attached brass or copper plate. Unless otherwise specified (see 6.2), suitable flanged elbows shall be furnished as required to permit installation of piping parallel to the axis of pump rotation. The pump-engine system shall be complete so that when installed, it can be used for any operation for which it was designed. The pump-engine system shall be free of any characteristics or defects that will prevent it from passing any of the inspections and tests specified in section 4.

3.7 Pump performance. Unless otherwise specified (see 6.2), the pump shall provide a flow rate of 350 gallons per minute (gpm) against a total dynamic head of 250 feet (ft), and a liquid suction head of 15 ft. The pump-engine system shall meet the performance requirements specified when tested in accordance with section 4 of this specification. The pump performance shall be met with the engine speed controls set at an adjustment not exceeding that producing the net continuous brake horsepower (bhp). To preclude the use of an overpowered pump-engine system, the test shall be met but not exceeded by more than 20 percent while the engine speed controls are set at an adjustment not less than that producing 60 percent of the net continuous bhp. The pump efficiency shall be not less than 60 percent in meeting the specified performance requirements. The design shall be such that the pump shall not cavitate or tend to cavitate while pumping fresh water at 100 degrees Fahrenheit (°F) at sea-level conditions.

3.8 Diesel engine. The diesel engine shall be a standard 2- or 4-cycle engine. The continuous horsepower output-shaft rating shall be sufficient to meet the input requirements of the pump without exceeding the engine manufacturer’s published continuous bhp rating for the speed at which the engine will normally operate. The engine shall be furnished complete with all accessories normally supplied as standard equipment including a governor, oil filter, air cleaner, muffler, and fuel tank.

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3.8.1 Diesel starting system. A 12 or 24 volt starting system shall be furnished. A suitable, readily visible instruction plate of copper or brass shall be securely fastened to the engine housing. The plate shall outline the step-by-step procedure that insures safety and ease of starting. The equipment shall include an ether starting system or glow plug. When an ether priming system is required, it shall be of the measured shot type with reservoir capacity of not less than 12 fluid ounces. The ether system shall be key operated or manually operated from the instrument panel and shall be inoperative with the engine warm.

3.9 Engine components and accessories.

3.9.1 Exhaust system. The exhaust system and muffler shall be the manufacturer's standard design. When a commercial-type ejector pump primer is furnished (see 3.10.2), modifications which supply sufficient volume and pressure of exhaust gas, without damage to engine, muffler, or exhaust system is acceptable.

3.9.2 Battery(s). Each battery shall be of 12 volt potential. The reserve capacity for each battery shall be not less than 110 minutes and the cold cranking rating at 0°F shall be not less than 450 amperes, measured in accordance with SAE J537. Unless otherwise specified (see 6.2), battery(s) shall be maintenance-free having the maintenance-free characteristics listed in W-B-131. When specified (see 6.2), battery(s) shall be dry-charged in accordance with W-B-131, without electrolyte, with sealed caps to prevent the intrusion of atmospheric moisture.

3.9.3 Charging system. The battery charging generator/alternator and regulators shall be the manufacturer's standard type, and in addition to its primary function, shall have the capacity sufficient to support a specified auxiliary lighting system of 120 watts (see 3.10.3).

3.9.4 Fuel system. The fuel system shall be provided with a 3-way sealed-lug type selector valve and an external auxiliary connection that has a protective cap secured against loss. The external auxiliary connection shall be a SAE standard male, flared fitting for mating with the auxiliary fuel supply. A circular faceplate marked "OFF", "UNIT TANK", and "AUXILIARY", to indicate the flow of fuel from the engine, valve shall be provided. The fuel tank shall, without recourse to the auxiliary source, have capacity for continuous operation for a period of not less than 6 hours under normal load.

3.9.5 Instrument panel. An instrument panel shall be furnished, equipped with the following devices:

a. Gage, pressure, lubricating oil.
b. Tachometer.
c. Indicator, battery-generator.
d. Indicator, cooling liquid temperature.
e. Starter, pushbutton or other switch when engine is equipped with solenoid-type switch.
f. Key-operated switch for an ether diesel engine starting system, when required (see 3.8.1).
g. Key-operated switch for a glow plug diesel engine starting system, when required (see 3.8.1)

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h. Suitable connector (plug, jack, or terminal) for extension cord type portable lighting system to be furnished when auxiliary lighting is specified (see 3.10.3).
i. Suitably located incandescent bulbs for indirect lighting of instrument panel.

3.9.6 Engine hour meter. An engine hour meter indicating total operation time shall be provided which registers up to 9,999.9 hours in 0.1 hour increments.

3.10 Pump components and accessories.

3.10.1 Pressure gage. A compound gage reading from O to 30 inches of mercury vacuum and O to 150 pound-force per square inch gage (psig) shall be furnished for the pump intake. A pressure gage in a range of O to 200 psig shall be furnished for the pump discharge. Both gages shall be furnished with pulsation dampeners and gage clocks. Dials shall have a minimum nominal diameter of not less than 2-1/2 inches. Gages shall be of the Bourdon-type and shall conform to the requirements of ASME B 40.1.

3.10.1.1 Gage savers. Gage savers shall be provided on the lines connecting the pressure gages and safety switches to the pump to protect then from damage caused by freezing. The device or devices used for pressure-dampening, liquid barriers, and the isolated liquid between the barriers and the gages shall assure satisfactory operation of the instruments under all operating conditions of the pump.

3.10.2 Auxiliary priming system. Unless otherwise specified (see 6,2), a dual system, auxiliary priming system shall be furnished. One component of the system shall be a commercially available hand pump capable of filling the centrifugal pump and suction line with primer liquid. The other component shall be a manually-controlled exhaust ejector, or a manually-controlled suction device engine to prime the pump. Automatic means, by float or otherwise, shall be provided to prevent entrance of liquid into the intake manifold during primer operation. Means shall be provided for isolating the components from the centrifugal pump and for draining each component of the system so that drainage goes back into the volute or other areas after operation. A readily visible brass or copper plate securely fastened in the vicinity of each of the two priming devices shall outline (step-by-step) the operating procedure for each. The plate shall include a warning against running the engine to accomplish priming of a dry pump by use of the ejector or vacuum system except for repriming purposes in which the residual liquid of the volute can act as a lubricant. When an ejector primer is furnished, the instructions shall specify that the ejector shall not be operated longer than 2 minutes at a time, and that the system shall be dismantled and cleaned after each 200 hours of engine operation. Instructions shall indicate how to completely isolate the ejector system when flammable liquids are pumped.

3.10.3 Auxiliary lighting. When specified (see 6.2), an auxiliary lighting system for night operation shall be provided. The system shall consist of two separate units consisting of bulb, lens-enclosed reflector, spring-actuated clamp or magnetic clamp for reflector, 20-foot extension cord with jack plug, or terminal leads for connection on instrument panel. The cord shall be in accordance with NFPA 70, type SO, for use in damp places, and comprised of

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3-wire conductors of a size sufficient to safely carry a current twice the lamp amperage. The bulb shall be of 40 to 60 watt capacity, shall have voltage compatible with a generator-battery system, and shall be of a standard bayonet-socket type.

3.10.4 Line pressure safety switch system. The pump-engine system shall be equipped with a safety switch system for immediate and positive throttling of the engine to idle speed. The system shall consist of the following components: a hermetically-sealed, pressure-actuated, electric high-pressure switch and low-pressure switch, solenoid, toggle switch with indicating light, and independent electric circuit with voltage compatible with cranking system. The safety switches shall be integrated with the engine governor system and shall control the action of the fuel injection pump. Any one or a combination of the following high-and low-pressure conditions shall cause the safety-switch system to function.

a. A drop in pressure below the preset limits when the flow of liquid in the upstream (suction) line is cut off, leaving the pump to operate dry.
b. A drop in pressure below the preset limits when the liquid is being wasted because of a break in the downstream (discharge) line.
c. A rise in the pressure beyond the preset limits due to a block in the downstream (discharge) line.

3.10.4.1 Pressure-actuated switches. The high- and low-pressure safety switches shall be exterior, surface-mounted, bellows-type, pressure-actuated, normally open switches with positive, quick-make-and-break action. The operating range of the high-pressure switch shall be 90 to 900 pound-force per square inch (psi) plus or minus 5 percent; the operating range of the low-pressure switch shall be 10 to 500 psi plus or minus 5 percent. The switches shall be adjustable for any set point within their respective operating ranges, The switches shall be connected to the pump discharge nozzle with suitable copper tubing and fittings.

3.10.4.2 Solenoid. A weather-proof, pull-type, manual-resetting solenoid shall be securely mounted on the engine with the plunger arm connected to the diesel fuel injection pump. The solenoid shall be connected by electric cables to the rest of the control system. When energized, the plunger arm shall act to have the engine operate at idling speed. Means shall be provided for adjusting the engine idling speed to the recommended speed when the solenoid is energized.

3.10.4.3 Electric circuit. The wiring shall be arranged so that operation of the toggle switch will open or close the circuit to the solenoid. When the toggle switch is in the ON position, the circuit is closed and the indicating green light shall be energized. When the light is out, this shall indicate the high- and low-pressure switches have been cut out and that the toggle switch must be reset before the pressure-actuated safety system functions again. When the pump-engine system is put into operation and the line pressure is below the preset low limit, opening of the toggle switch shall permit cutting out the pressure switches and allow the engine to start.

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3.11 Skid base. The pump and engine shall be mounted on a common steel skid base, properly braced and reinforced by crossmembers to adequately support the fully assembled system and hold the pump and engine in proper alignment, particularly during times when it is towed along the ground or handled during shipment. Each runner of the skid base shall have both ends upturned in a manner to facilitate skidding. Provision shall be made for bolting the base to a permanent foundation by uniformly space holes in the skids, by nonprojecting lugs, or by other means. Means shall also be provided to permit lifting of the system with lifting slings. Such means may be 3-inch holes at both ends of each skid runner or welded lifting attachments. In either case, when the system is hoisted, there shall be adequate clearance between slings and exterior parts of the equipment. Holes or lifting attachments shall be so located that when hoisted, the system is in its operating attitude and no fuel, lubricating oil, cooling liquid, battery liquid, or other material shall spill during handling. The four holes or lifting attachments shall be capable of supporting 2-1/2 times the full weight of an operable system.

3.12 Lubrication. Unless otherwise specified (see 6.2), means for lubricating shall be provided in accordance with the manufacturer's standard practice. Parts requiring lubrication shall be so located as to make the lubricating points easily visible and accessible. All parts requiring lubrication shall be properly lubricated before delivery. Manufacturer's standard means for pressure lubrication shall be provided.

3.13 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion products or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 roils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.14 Dissimilar metals. Metals dissimilar with respect to the galvanic scale shall not be used unless separated by an insulating material which will avoid electrolytic corrosion.

3.15 Workmanship.

3.15.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

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3.15.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.15.3 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.15.4 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

3.15.5 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract, Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4 2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. Packaging inspection (see 4.5).

4.2.1 First article inspection. First article inspection shall be performed on one pump system when a first article is required (see 3.2 and 6.6) . This inspection shall include the examination of 4.3 and the tests of 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall consist of the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5. When a first article is tested in accordance with the requirements of 4.4, and has passed, the tests of 4.4 need not be performed for the quality conformance inspection. When the tests of 4.4 are to be performed for the quality conformance inspection, they shall be performed on one production unit.

4.3 Examination. Each engine-pump system hall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article, when furnished, shall be tested, and any system failing to pass the following tests, as applicable, shall be rejected. Tests shall be conducted as outlined in the referenced documents and as herein specified.

4.4.1 Performance test. The pump-engine system shall be operated continuously using water for period of not less than four hours. After the four-hour test, the pump-engine system shall be run continuously for 30 minutes through a range of speeds obtained with maximum-to-minimum engine speed control settings. No critical speeds shall be permitted at any point of the range. No leakage shall be evident in the casing or through the casing gasket. Minor leakage through the stuffing boxes or mechanical seals shall not be cause for rejection.

4.4.2 Hydrostatic test. The pump casing shall be subjected to a hydrostatic test, with the pressure equal to 1.5 times the maximum design pressure developed at full throttle operation, in accordance with the Standards of Hydraulic Institute and ASME B 73.1M or API STD 610, as applicable.

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4.5 Packaging inspection. The preservation, packing, and marking of the pump-engine system shall be inspected to the requirements of section 4 of MIL-P-10603. When specified (see 6.2), a preproduction pack inspection shall be performed in accordance with MIL-P-10603.

5. PACKAGING

5.1 Preservation, packing, and marking The pump-engine system shall be packaged, packed, and marked in accordance with MIL-P-10603. The level of preservation and level of packing shall be as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pump-engine systems are intended for applications at temporary and semi-permanent installations for handling fresh or salt water, gasoline, fuel oil, or diesel oil.

6.1.1 Selection factors.

a. Fungus resistant varnish conforming to MIL-V-173 should be used to coat electrical components and circuit elements, including terminal and circuit connections, when the system is to be installed in humid conditions. Components and elements inherently inert to fungi or in hermetically sealed enclosures or current-carrying contact surfaces should not be coated.
b. Electromagnetic interference suppression (EMI), when required, should conform to the EMI suppression requirements and test limits for class C3, group I equipment as specified in MIL-STD-461. The system should be subjected to tests to determine conformance in accordance with MIL-STD-462.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When either horizontal or vertical splitting of casing is specified (see 3.1).
d. When a first article is required (see 3.2 and 6.6).
e. Size of intake and discharge connection, if different (see 3.6).
f. If flanged elbows are not required (see 3.6).
g. When major performance requirements are other than as stated (see 3.7); flow in gpm, total dynamic head in feet, and liquid suction head in feet.
h. When battery(s) is not to be maintenance free (see 3.9.2).
i. When dry-charged type battery(s) is required (see 3.9.2).
j. When an auxiliary priming system is not required (see 3.10.2).
k. When an auxiliary lighting system is not required (see 3.10.3).

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l. When other than manufacturer's standard practice of lubrication required (see 3.12).
m. When the preproduction pack inspection requirements are required (see 4.5).
n. Level of preservation and level of packing required (see 5.1).

6.3 Part or identifying number (PIN). The PIN to be used for systems applied to this specification are created as follows:

	M17574	2
Specification part number		
Type code number		

6.4 Cross-reference of classification. The classification of pump-engine systems in this document differs from the previous revision in the following respect:

MIL-P-17574C	MIL-P-17574D
TYPES	
Type I - Gasoline engine-driven.	Type I - Deleted.
Type II - Diesel engine-driven.	Type II - Diesel-engine-driven.

6.5 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27475.1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data will be delivered by the contractor in accordance with the contract or purchase order requirements.

6.6 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory (see 6.2) as specified in 4.2.1. The first article should consist of one system. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.7 Subject term (key word) listing.

Nonsubmersible
Diesel engine

6.8 Changes from Previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:
Navy - YD

(Project 4320-N315)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-17574D(YD)	2. DOCUMENT DATE (YYMMDD) 910621

3. DOCUMENT TITLE
PUMP-ENGINE-SYSTEM, CENTRIFUGAL, WATER-GASOLINE-FUEL OIL, SKID-BASE-MOUNTED

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDING OFFICER	b. TELEPHONE (Include Area Code) (1) Commercial: (805) 982-5761 (2) AUTOVON: 551-5761
c. ADDRESS (Include Zip Code) NAVAL CONSTRUCTION BATTALION CENTER CESO (CODE 1564A) PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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